

EXPERIMENT 6

```
mysql> show databases;
```

Database
csa050
csa0508
dbms
exp1
information_schema
mysql
new
performance_schema
sailors
sys

```
10 rows in set (0.03 sec)
```

```
mysql> create database exp6;
```

```
Query OK, 1 row affected (0.01 sec)
```

```
mysql> use exp6;
```

```
Database changed
```

```
mysql> CREATE TABLE DEPT (
```

```
->     DEPTNO INT PRIMARY KEY,
```

```
->     DNAME VARCHAR(20),
```

```
->     LOC VARCHAR(20)
```

```
-> );
```

```
Query OK, 0 rows affected (0.05 sec)
```

```
mysql> INSERT INTO DEPT VALUES
```

```
-> (10, 'ACCOUNTING', 'NEW YORK'),
```

```
-> (20, 'RESEARCH', 'DALLAS'),
```

```
-> (30, 'SALES', 'CHICAGO'),
```

```
-> (40, 'OPERATIONS', 'BOSTON');
```

```
Query OK, 4 rows affected (0.02 sec)
```

```
Records: 4  Duplicates: 0  Warnings: 0
```

```
mysql> select * from exp6;
```

```
ERROR 1146 (42S02): Table 'exp6.exp6' doesn't exist
```

```
mysql> select * from dept;
```

DEPTNO	DNAME	LOC
10	ACCOUNTING	NEW YORK
20	RESEARCH	DALLAS
30	SALES	CHICAGO
40	OPERATIONS	BOSTON

4 rows in set (0.00 sec)

```
mysql> CREATE TABLE EMP (  
-> EMPNO INT PRIMARY KEY,  
-> ENAME VARCHAR(20),  
-> JOB VARCHAR(20),  
-> MGR INT,  
-> HIREDATE DATE,  
-> SAL DECIMAL(10,2),  
-> COMM DECIMAL(10,2),  
-> DEPTNO INT,  
-> FOREIGN KEY (DEPTNO) REFERENCES DEPT(DEPTNO)  
-> );
```

Query OK, 0 rows affected (0.06 sec)

```
mysql> select * from emp;  
Empty set (0.00 sec)
```

```
mysql> INSERT INTO EMP VALUES  
-> (7369, 'SMITH', 'CLERK', 7902, '1980-12-17', 800, NULL, 20),  
-> (7499, 'ALLEN', 'SALESMAN', 7698, '1981-02-20', 1600, 300, 30),  
-> (7521, 'WARD', 'SALESMAN', 7698, '1981-02-22', 1250, 500, 30),  
-> (7566, 'JONES', 'MANAGER', 7839, '1981-04-02', 2975, NULL, 20),  
-> (7654, 'MARTIN', 'SALESMAN', 7698, '1981-09-28', 1250, 1400, 30),  
-> (7698, 'BLAKE', 'MANAGER', 7839, '1981-05-01', 2850, NULL, 30),  
-> (7782, 'CLARK', 'MANAGER', 7839, '1981-06-09', 2450, NULL, 10),  
-> (7788, 'SCOTT', 'ANALYST', 7566, '1982-12-09', 3000, NULL, 20),  
-> (7839, 'KING', 'PRESIDENT', NULL, '1981-11-17', 5000, NULL, 10),  
-> (7844, 'TURNER', 'SALESMAN', 7698, '1981-09-08', 1500, 0, 30),  
-> (7876, 'ADAMS', 'CLERK', 7788, '1983-01-12', 1100, NULL, 20),  
-> (7900, 'JAMES', 'CLERK', 7698, '1981-12-03', 950, NULL, 30),  
-> (7902, 'FORD', 'ANALYST', 7566, '1981-12-03', 3000, NULL, 20),  
-> (7934, 'MILLER', 'CLERK', 7782, '1982-01-23', 1300, NULL, 10);
```

Query OK, 14 rows affected (0.03 sec)

Records: 14 Duplicates: 0 Warnings: 0

```
mysql> select * from emp;
```

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
7369	SMITH	CLERK	7902	1980-12-17	800.00	NULL	20
7499	ALLEN	SALESMAN	7698	1981-02-20	1600.00	300.00	30
7521	WARD	SALESMAN	7698	1981-02-22	1250.00	500.00	30
7566	JONES	MANAGER	7839	1981-04-02	2975.00	NULL	20
7654	MARTIN	SALESMAN	7698	1981-09-28	1250.00	1400.00	30
7698	BLAKE	MANAGER	7839	1981-05-01	2850.00	NULL	30
7782	CLARK	MANAGER	7839	1981-06-09	2450.00	NULL	10
7788	SCOTT	ANALYST	7566	1982-12-09	3000.00	NULL	20
7839	KING	PRESIDENT	NULL	1981-11-17	5000.00	NULL	10
7844	TURNER	SALESMAN	7698	1981-09-08	1500.00	0.00	30

7876	ADAMS	CLERK	7788	1983-01-12	1100.00	NULL	20
7900	JAMES	CLERK	7698	1981-12-03	950.00	NULL	30
7902	FORD	ANALYST	7566	1981-12-03	3000.00	NULL	20
7934	MILLER	CLERK	7782	1982-01-23	1300.00	NULL	10

14 rows in set (0.00 sec)

mysql> UPDATE EMP

-> SET SAL = 18000

-> WHERE EMPNO = 7839;

Query OK, 1 row affected (0.02 sec)

Rows matched: 1 Changed: 1 Warnings: 0

mysql>

mysql> UPDATE EMP

-> SET SAL = 17000

-> WHERE EMPNO = 7698;

Query OK, 1 row affected (0.02 sec)

Rows matched: 1 Changed: 1 Warnings: 0

mysql> select * from emp;

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
7369	SMITH	CLERK	7902	1980-12-17	800.00	NULL	20
7499	ALLEN	SALESMAN	7698	1981-02-20	1600.00	300.00	30
7521	WARD	SALESMAN	7698	1981-02-22	1250.00	500.00	30
7566	JONES	MANAGER	7839	1981-04-02	2975.00	NULL	20
7654	MARTIN	SALESMAN	7698	1981-09-28	1250.00	1400.00	30
7698	BLAKE	MANAGER	7839	1981-05-01	17000.00	NULL	30
7782	CLARK	MANAGER	7839	1981-06-09	2450.00	NULL	10
7788	SCOTT	ANALYST	7566	1982-12-09	3000.00	NULL	20
7839	KING	PRESIDENT	NULL	1981-11-17	18000.00	NULL	10
7844	TURNER	SALESMAN	7698	1981-09-08	1500.00	0.00	30
7876	ADAMS	CLERK	7788	1983-01-12	1100.00	NULL	20
7900	JAMES	CLERK	7698	1981-12-03	950.00	NULL	30
7902	FORD	ANALYST	7566	1981-12-03	3000.00	NULL	20
7934	MILLER	CLERK	7782	1982-01-23	1300.00	NULL	10

14 rows in set (0.00 sec)

mysql> SELECT JOB, SUM(SAL) AS TOTAL_SALARY

-> FROM EMP

-> GROUP BY JOB;

JOB	TOTAL_SALARY
CLERK	4150.00
SALESMAN	5600.00
MANAGER	22425.00

ANALYST	6000.00
PRESIDENT	18000.00

5 rows in set (0.00 sec)

```
mysql> SELECT *
-> FROM EMP E1
-> WHERE SAL = (
->     SELECT MIN(SAL)
->     FROM EMP E2
->     WHERE E1.MGR = E2.MGR
-> );
```

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
7369	SMITH	CLERK	7902	1980-12-17	800.00	NULL	20
7782	CLARK	MANAGER	7839	1981-06-09	2450.00	NULL	10
7788	SCOTT	ANALYST	7566	1982-12-09	3000.00	NULL	20
7876	ADAMS	CLERK	7788	1983-01-12	1100.00	NULL	20
7900	JAMES	CLERK	7698	1981-12-03	950.00	NULL	30
7902	FORD	ANALYST	7566	1981-12-03	3000.00	NULL	20
7934	MILLER	CLERK	7782	1982-01-23	1300.00	NULL	10

7 rows in set (0.00 sec)

```
mysql> SELECT D.DEPTNO, D.DNAME, COUNT(E.EMPNO) AS NO_OF_EMPLOYEES
-> FROM DEPT D
-> JOIN EMP E
-> ON D.DEPTNO = E.DEPTNO
-> GROUP BY D.DEPTNO, D.DNAME;
```

DEPTNO	DNAME	NO_OF_EMPLOYEES
10	ACCOUNTING	3
20	RESEARCH	5
30	SALES	6

3 rows in set (0.00 sec)

```
mysql> SELECT *
-> FROM EMP
-> ORDER BY SAL ASC;
```

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
7369	SMITH	CLERK	7902	1980-12-17	800.00	NULL	20
7900	JAMES	CLERK	7698	1981-12-03	950.00	NULL	30
7876	ADAMS	CLERK	7788	1983-01-12	1100.00	NULL	20
7521	WARD	SALESMAN	7698	1981-02-22	1250.00	500.00	30
7654	MARTIN	SALESMAN	7698	1981-09-28	1250.00	1400.00	30

7934	MILLER	CLERK	7782	1982-01-23	1300.00	NULL	10
7844	TURNER	SALESMAN	7698	1981-09-08	1500.00	0.00	30
7499	ALLEN	SALESMAN	7698	1981-02-20	1600.00	300.00	30
7782	CLARK	MANAGER	7839	1981-06-09	2450.00	NULL	10
7566	JONES	MANAGER	7839	1981-04-02	2975.00	NULL	20
7788	SCOTT	ANALYST	7566	1982-12-09	3000.00	NULL	20
7902	FORD	ANALYST	7566	1981-12-03	3000.00	NULL	20
7698	BLAKE	MANAGER	7839	1981-05-01	17000.00	NULL	30
7839	KING	PRESIDENT	NULL	1981-11-17	18000.00	NULL	10

14 rows in set (0.00 sec)

```
mysql> SELECT *
-> FROM EMP
-> WHERE SAL > 16000
-> ORDER BY DEPTNO;
```

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
7839	KING	PRESIDENT	NULL	1981-11-17	18000.00	NULL	10
7698	BLAKE	MANAGER	7839	1981-05-01	17000.00	NULL	30

2 rows in set (0.00 sec)